Engine Model(s): PT6A

73-20-00

FUEL CONTROL UNIT - DESCRIPTION AND OPERATION

1. Description

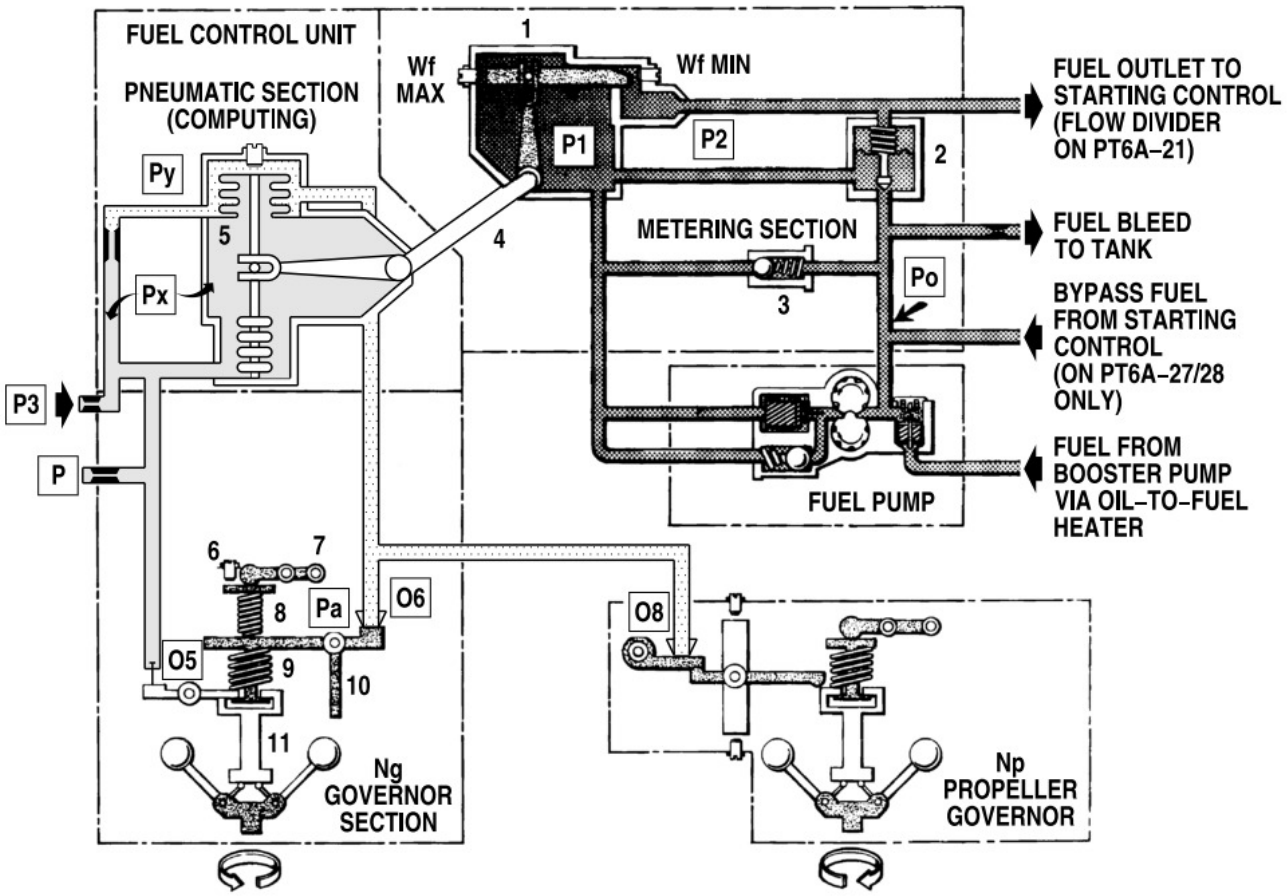
A. General

The fuel control unit (FCU) is mounted on the rear flange of the fuel pump. A splined coupling between the pump and the FCU transmits a speed signal, proportional to compressor turbine speed (Ng), to the governing section in the FCU. The FCU determines the fuel schedule for the engine to provide the power required as established by controlling the speed of the compressor. Engine power output is directly dependent upon Ng. Control of the Ng is done by regulating the amount of fuel to the combustion section of the engine.

2. Operation (Ref. Fig. 1)

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Figure 1 Fuel Control Unit Schematic



LEGEND

1. FUEL METERING VALVE
2. BYPASS VALVE
3. HIGH PRESSURE RELIEF VALVE
4. TORQUE TUBE ASSEMBLY
5. ROD
6. IDLE SPEED ADJUSTMENT
7. SPEED SET LEVER
8. GOVERNOR SPRING
9. ENRICHMENT SPRING
10. LINK
11. COMPRESSOR TURBINE GOVERNOR PLATFORM

Pa AMBIENT AIR
P1 UNMETERED PUMP DELIVERY FUEL
P2 METERED FUEL
P3 COMPRESSOR DISCHARGE AIR
Po BYPASS FUEL
Px ENRICHMENT PRESSURE
Py GOVERNING PRESSURE
O5 ENRICHMENT ORIFICE
O6 GOBERNING ORIFICE
O8 PROPELLER GOVERNOR Nf SECTION ORIFICE

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A. Metering Section

The FCU is supplied with fuel at pump pressure (P1). Fuel flow is established by a metering valve and bypass valve system, with P1 fuel supplied to the metering valve input. The fuel pressure immediately after the metering valve is called metered fuel (P2), which flows to the starting control unit. The bypass valve maintains an essentially constant fuel pressure differential (P1 minus P2) across the metering valve. The orifice area of the metering valve is changed by valve movement to meet specific engine requirements, with fuel pump output (P1) in excess of requirements returned, via internal passages in the FCU and pump, to the pump inlet downstream of inlet screen. Bypassed fuel is referred to as Po fuel. The bypass valve consists of a sliding valve operating in a ported sleeve and is actuated via a diaphragm and spring. In operation, the spring force is balanced by the P1 minus P2 differential operating on the diaphragm. The valve is always in a position to maintain the P1 minus P2 difference, and to bypass Po fuel in excess of engine requirements.

A hign pressure relief valve is incorporated in parallet with the bypass valve to prevent excessive P1 pressure in the FCU. The valve is spring loaded closed and remains closed until the inlet pressure (P1) overcomes the spring force and opens the valve to Po. As soon as the pressure is reduced to an acceptable value, the valve closes.

The metering valve consists of a contoured needle operating in a sleeve, and regulating the flow of fuel by varying the orifice area. Fuel flow after initial start is a function of metering valve position only because the bypass valve maintains an essentially constant differential pressure across the orifice regardless of fuel input or discharge pressure variations.

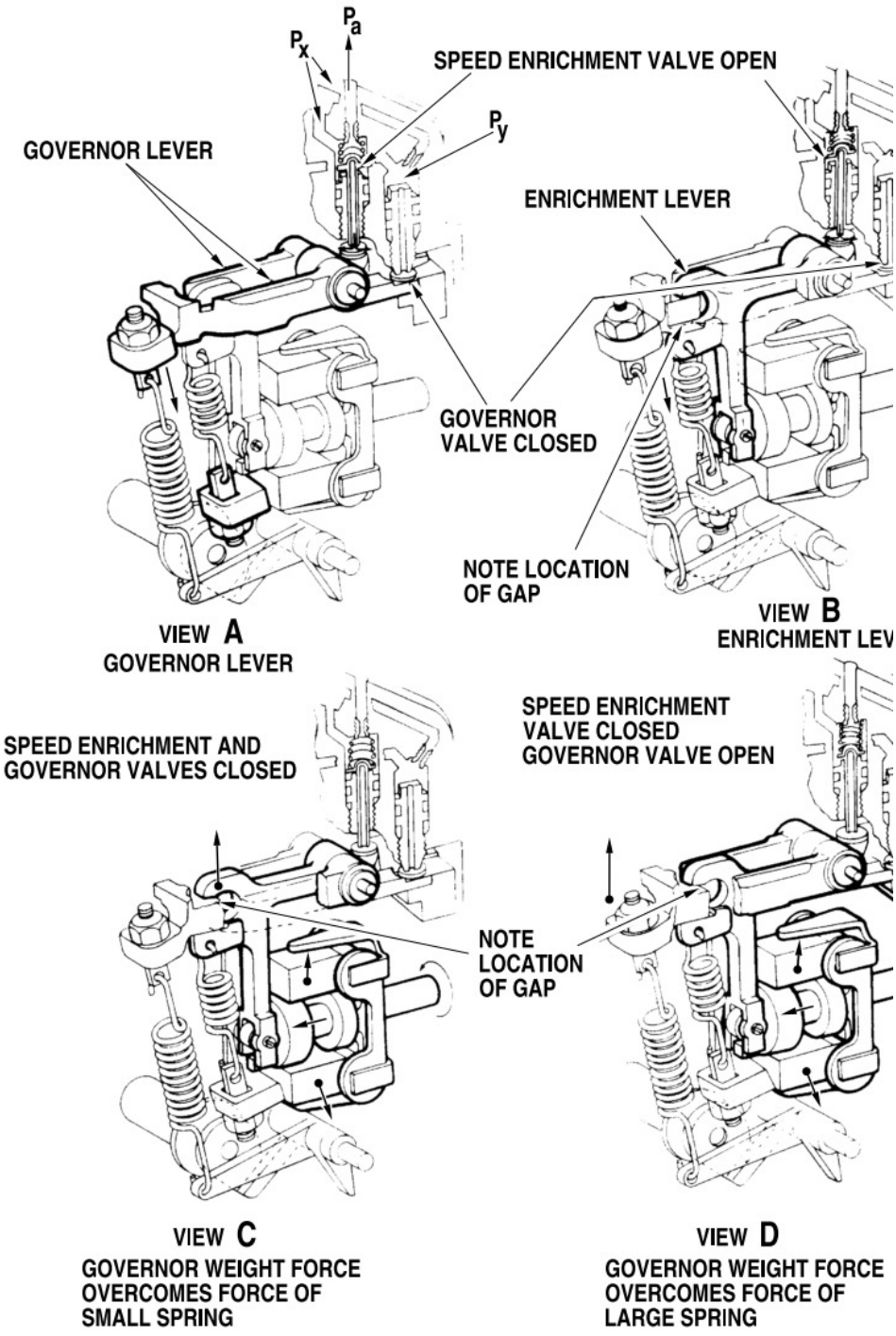
An external adjustment is provided on the bypass valve spring cover to permit variation of acceleration rate, and acceleration matching, on multi-engined installations.

Variations in specific gravity of the fuel resulting from changes in fuel temperature is compensated for with bimetallic disks installed under the bypass valve spring.

B. Power Input and Speed Governing Section (Ref. Figs. 1 and 2)

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Figure 2 FCU/Governor Drive Body Assembly – Operation



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The enrichment lever provides a mechanical link between the governor spool and governor lever. The power input shaft incorporates a speed scheduling cam which depresses a cam follower lever when power is increased. A spring connects the cam follower lever to the governor lever. The governor lever is pivoted and one end operates against an airflow restrictor to form the governor valve. The enrichment lever pivots at the same point as the governor lever and has two extensions which straddle a portion of the governor lever so that after a slight movement, a gap closes and both levers move together (Ref. Fig. 2, View A and View B). A smaller spring connects the enrichment lever to the governor lever. A roller on the arm of the enrichment lever contacts the end of the governor spool.

The speed scheduling cam applies tension to the governor spring through an intermediate lever/cam roller linkage which applies a force to close the governor valve.

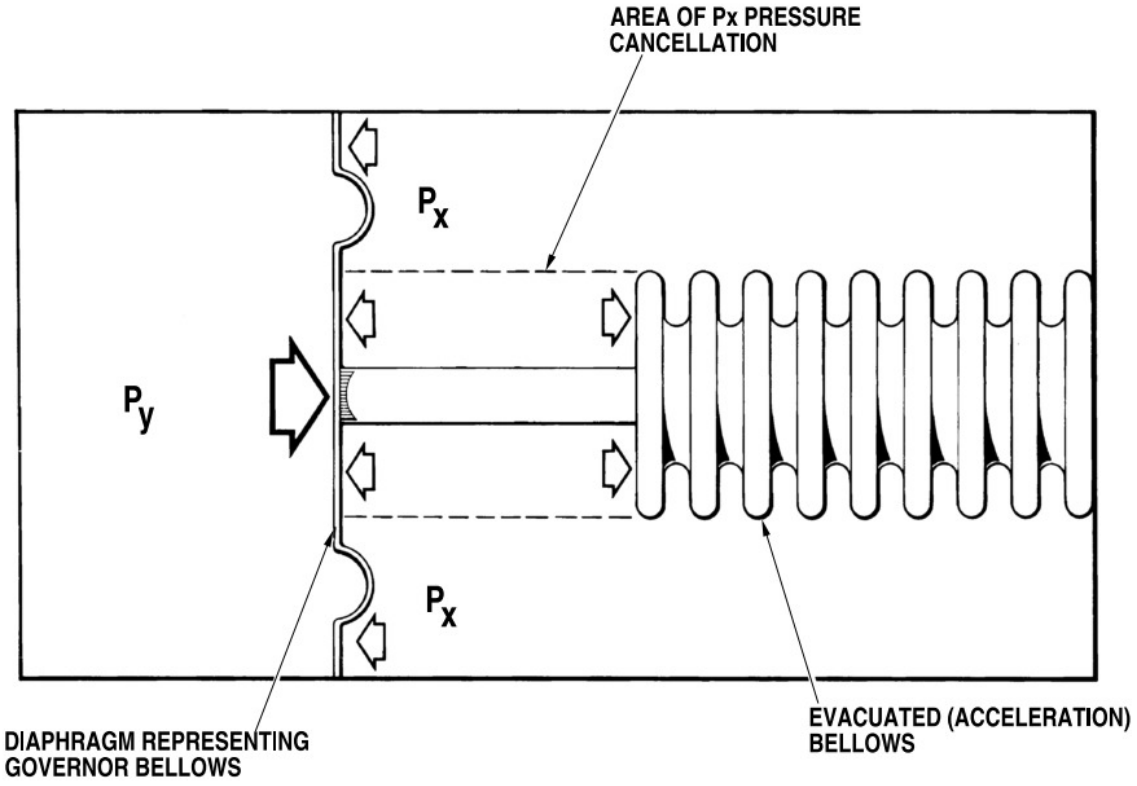
As the drive shaft revolves, it rotates a table to which the governor weights are attached. Small levers on the inner face of the weights contact the base of the governor spool. As the Ng increases, the drive shaft rotates faster; this causes the weights to pivot outward under centrifugal force, which in turn causes the small levers to exert an increasing force on the spool. This moves the spool outward on its shaft to bear against the enrichment lever roller. As governor weight force overcomes opposing spring force, the governor valve is opened. The enrichment lever will start to move whenever the Ng increases enough to cause the weight force to overcome the force of the smaller spring, until the lever contacts the governor lever (Ref. Fig. 2, View C). The governor valve will open if the Ng increases sufficiently to cause the weight force to overcome the force of the larger spring. At this point, the governor valve will be open (Ref. Fig. 2, View D).

A drain hole is incorporated to vent the inner cavity to ambient atmospheric pressure (Pa). Modified compressor discharge pressure (Px) is bled off to the atmosphere when the governor valve is open.

C. Computing Section (Ref. Figs. 1 and 3)

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Figure 3 FCU Computing Section Functional Diagram



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The computing section consists of an evacuated (acceleration) bellows and a governing bellows connected by a common rod. The end of the acceleration bellows, opposite the rod, is attached to the body casting. The acceleration bellows provides an absolute pressure reference. The governor bellows is secured in the body cavity and its function is similar to that of a diaphragm. Movement of the bellows is transmitted to the metering valve by the cross shaft and associated levers. The cross shaft moves within a torque tube assembly which is attached to the cross shaft near the bellows lever. The torque tube is secured in the body casting at the opposite end by an adjustment bushing. Therefore, any rotational movement of the cross shaft will result in an increase and decrease in the force of the torque tube. The torque tube forms the seal between the computing and metering sections of the FCU. The tube is positioned during assembly to provide a force in a direction tending to close the metering valve while the governor bellows act against this force to open the valve. Py pressure is applied to the outside of the governor bellows, while Px pressure is applied to the inside of the bellows and to the outside of the acceleration bellows.

For explanation purposes (Ref. Fig. 3), the governor bellows is shown as a diaphragm. Py pressure is applied to one side of the "diaphragm" and Px pressure is applied to the other side. Px pressure is also applied to the outside of the acceleration bellows which is also attached to the diaphragm. The force of Px applied against the acceleration bellows is cancelled by application of the same pressure on an equal area of the diaphragm, as the forces act in opposite directions. All pressure forces applied to the computing section can be resolved into forces acting on the diaphragm only. The forces are:

1. Py pressure acting on the entire surface on one side.
1. Px pressure acting on the opposite surface partially cancelled by the effect of Px acting on the face of the evacuated bellows connected to the governor bellows.

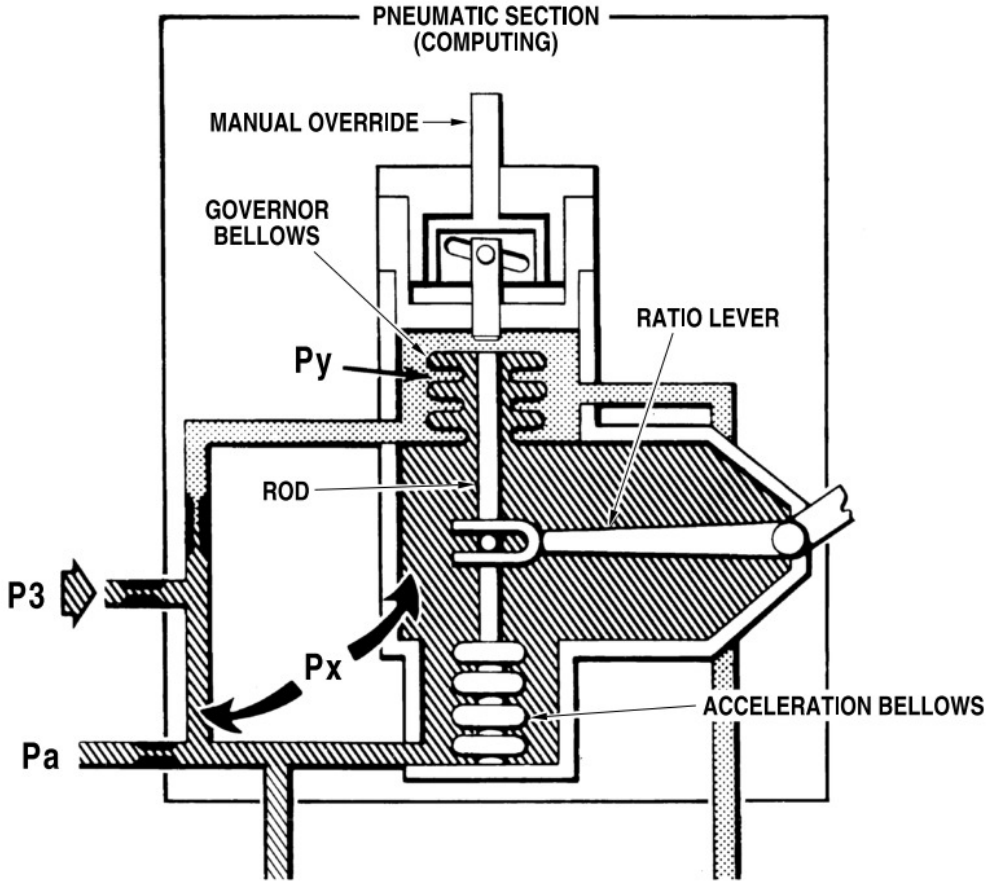
Any change in Py will therefore have more effect on the diaphragm than an equal change in Px pressure, due to difference in effective area.

Px and Py vary with changing engine operating conditions as well as inlet air temperature. When both pressures increase simultaneously, as during acceleration, the bellows cause the metering valve to move in an opening direction. When Py decreases as the desired Ng is approached (for governing after acceleration), the bellows will travel to reduce the opening of the metering valve. When both pressures decrease simultaneously, the bellows will travel to reduce the metering valve opening because a change in Py is more effective than the same change in Px. This occurs during deceleration and moves the metering valve to its minimum flow stop.

D. Manual Override System (Post-SB1469, PT6A-27 Engines) (Ref. Fig. 4)

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Figure 4 Manual Override System



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On FCU's incorporating the manual override system, the retaining plate and cover containing the governor bellows stop are replaced by a shaft and stop assembly, which, if operated after a pneumatic failure, pushes against the end of the governor bellows to increase fuel flow.

The shaft and stop assembly consists of an actuating shaft incorporating spiral slots; a driving pin fits in the slots, passing through and activating an operating pin. When the actuating shaft is turned, the driving pin, which cannot rotate, moves in or out dependant on the slot position. Subsequent movement of the perating pin against the end of governor bellows opens the metering valve and increases fuel flow. In the OFF position, the operating pin acts as bellows travel stop.

E. Power Turbine (Nf) Governor (Ref. Fig. 1)

The power turbine Nf governor section of the propeller governor (Ref. 61-20-00) senses Py pressure through an external pneumatic line from the computing section of the FCU to the Nf governor. In the event of a power turbine overspeed condition, a governing orifice (08) in the Nf governing section is opened by flyweight action of the governor to bleed off Py pressure through the governor to atmosphere. When this occurs, Py pressure acting on the FCU governor bellows decreases and moves the metering valve in a closing direction, thus reducing fuel flow; this in turn decreases Ng speed and consequently Nf speed. The speed at which the Nf governing orifice (08) opens is dependent on the setting of the propeller governor speed set control and the setting of the Nf reset arm.

Normally, the Nf governing orifice (08) is opened at six percent above propeller governing speed setting with the Nf governor reset arm at maximum position and approximately four percent under propeller governor speed setting at minimum position. In reverse thrust, the propeller reversing interconnect linkage resets the Nf governor reset arm to a setting below the propeller governor speed set lever setting. Power turbine (Nf) speed, and hence propeller speed, is then limited by the Nf governor. Power from the gas generator is reduced to allow a propeller speed approximately four percent under the speed set by the propeller governor.

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